

CAPE YOUNG, NU, Canada

WMO: 719390

Lat: 68.933N

Lon: 116.917W

Elev: 16

StdP: 101.13

Time Zone: -7.00 (NAM)

Period: 82-93

WBAN: 26114

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-40.9	-39.2	-40.8	0.1	-36.5	-39.6	0.1	-36.1	21.4	-24.3	19.0	-16.6	2.7	270	1.081

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.2	19.8	14.0	16.1	12.1	13.3	10.3	14.0	18.6	12.2	16.0	10.3	13.2	4.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.7	8.6	16.6	9.8	7.5	14.2	8.0	6.7	11.9	39.4	18.5	34.6	16.0	30.1	13.4	25.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.0	12.9	11.2	DB	-42.4	24.9	1.6	2.8	-43.5	26.9	-44.5	28.5	-45.4	30.1	-46.6	32.1	(4)
(5)			WB	-38.5	18.7	0.8	4.2	-39.2	21.7	-39.7	24.1	-40.1	26.5	-40.7	29.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	-12.2	-29.4	-29.5	-26.3	-19.1	-7.6	2.8	6.9	6.6	1.7	-7.2	-21.1	-26.0	
	DBStd	14.90	6.16	7.25	6.01	6.49	5.68	3.99	4.52	3.62	3.07	5.23	7.06	6.25	
	HDD10.0	8162	1220	1106	1125	873	546	219	121	120	249	534	932	1117	
	HDD18.3	11160	1478	1339	1383	1123	804	466	354	363	498	793	1182	1375	
	CDD10.0	44	0	0	0	0	0	4	26	15	0	0	0	0	
	CDD18.3	1	0	0	0	0	0	0	1	0	0	0	0	0	
	CDH23.3	17	0	0	0	0	0	0	15	2	0	0	0	0	
	CDH26.7	4	0	0	0	0	0	0	4	0	0	0	0	0	
Wind	WSAvg	5.0	5.2	5.3	5.1	4.4	4.3	4.5	4.6	5.0	5.6	5.9	4.8	5.0	
Precipitation	PrecAvg	177	4	4	4	7	7	16	34	35	28	19	11	5	
	PrecMax	282	12	16	15	47	27	46	86	80	74	65	33	22	
	PrecMin	85	0	0	0	0	0	1	2	5	7	1	0	0	
	PrecStd	50	3	5	4	9	7	12	21	19	16	14	8	5	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-8.7	-6.8	-8.1	-1.1	5.1	21.0	25.1	23.1	13.2	4.2	3.1	-5.9	
		MCWB	-9.1	-7.1	-8.6	-1.4	3.6	13.3	16.5	16.2	9.6	3.5	0.8	-6.2	
	2%	DB	-12.0	-11.0	-11.0	-4.2	2.7	15.3	20.9	18.0	9.2	2.9	-3.8	-10.0	
		MCWB	-12.2	-11.2	-11.2	-4.6	1.7	10.8	14.2	13.2	7.1	2.1	-4.4	-10.2	
	5%	DB	-16.8	-15.1	-13.6	-6.9	1.2	11.4	17.2	15.2	7.1	0.9	-7.7	-12.8	
		MCWB	-17.0	-15.4	-13.8	-7.2	0.6	8.3	12.5	11.9	5.9	0.3	-8.1	-13.1	
	10%	DB	-20.1	-18.9	-16.8	-9.1	0.1	8.8	14.2	12.2	5.9	-0.2	-11.0	-16.0	
		MCWB	-20.3	-19.2	-17.0	-9.4	-0.2	6.5	11.0	9.8	4.9	-0.6	-11.3	-16.2	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-8.8	-7.0	-8.5	-1.5	3.7	13.2	18.0	16.9	9.7	3.6	1.1	-6.3
			MCDB	-8.4	-6.7	-8.2	-1.2	4.9	19.7	24.4	21.8	13.4	4.1	2.7	-6.1
2%		WB	-11.9	-10.8	-11.1	-4.7	1.6	10.8	14.7	13.7	7.2	2.0	-4.6	-10.4	
		MCDB	-11.8	-10.7	-11.0	-4.3	2.3	15.2	19.7	16.9	8.7	2.7	-4.0	-10.3	
5%		WB	-16.3	-14.4	-13.7	-7.3	0.6	8.5	12.9	11.8	6.0	0.2	-7.9	-13.0	
		MCDB	-16.2	-14.2	-13.5	-7.1	1.1	11.4	17.3	14.9	7.0	0.6	-7.6	-12.8	
10%		WB	-19.7	-18.4	-16.7	-9.6	-0.2	6.3	10.8	10.0	4.8	-0.8	-11.3	-16.1	
		MCDB	-19.6	-18.3	-16.6	-9.3	0.2	8.3	14.1	12.2	5.6	-0.5	-11.0	-15.9	
Mean Daily Temperature Range	5% DB	MDBR	6.1	6.8	7.4	8.5	6.7	5.6	7.2	5.7	4.2	5.1	6.1	6.5	
		MCDBR	9.0	9.6	9.4	9.3	6.0	11.6	14.2	11.5	7.2	4.6	7.8	7.7	
		MCWBR	8.9	9.1	9.4	9.4	5.5	7.8	8.8	7.4	5.5	4.3	7.7	7.7	
	5% WB	MCDBR	9.1	8.9	9.4	9.2	6.0	11.3	13.8	10.7	6.5	4.2	7.8	7.6	
		MCWBR	9.1	8.9	9.4	9.2	5.6	7.9	9.3	7.4	5.2	4.0	7.7	7.6	
Clear-Sky Solar Irradiance	taub	0.160	0.212	0.248	0.294	0.335	0.330	0.352	0.341	0.300	0.243	0.157	N/A		
	taud	1.886	2.047	2.190	2.128	2.115	2.340	2.389	2.464	2.509	2.167	1.921	N/A		
	Ebn at Noon	70	572	792	843	842	866	830	800	745	514	60	0		
	Edh at Noon	5	43	74	108	125	104	96	80	58	38	4	0		
All-Sky Solar Radiation	RadAvg	0.02	0.48	2.10	4.52	6.36	6.36	5.26	3.45	1.70	0.48	0.07	0.00		
	RadStd	0.00	0.04	0.10	0.21	0.34	0.45	0.36	0.35	0.19	0.07	0.01	0.00		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page